1. INTRODUCTION

I am trying to understand mobility friction in New York City spatially. There exists an enormous amount of administrative data documenting different hazards and obstacles encountered when moving through the city, from potholes and broken streetlights to crashes and double-parked cars. Individually, these datasets answer relatively specific questions; I want to combine them to ask a broader one: where is it difficult to move through New York City, and why?

This project is inspired by NYC Vision Zero and its tools for visualizing where crashes happen across the city, but the question is fundamentally different. Rather than asking how dangerous a street is, I am interested in how difficult it is to move through it. And that answer likely depends on who is moving: a pothole, broken bike rack, or double-parked car matters differently to a pedestrian, cyclist, or driver. The goal of this project is to develop and visualize measures of mobility friction for different modes of transportation through an interactive dashboard. I am especially interested in what patterns emerge across the city: where does friction concentrate, what produces it, and which communities experience the most?

2. LITERATURE REVIEW

As an economics major, it is important to me that the purpose of this paper is backed by theory on why mobility is important. The question begins with why people choose to live in cities. There are production advantages—high-paying jobs, thick labor markets, knowledge spillovers—as well as residential amenities—parks, hospitals, restaurants, and karaoke bars. Transportation functions alongside both: easy mobility lowers transportation costs for firms and expands the number of jobs and amenities residents can reach in the same amount of time.

An effective literature review, to me, therefore begins with a description across the field of urban economics of why mobility matters. And not just sources that explicitly advocate for the importance of mobility, but ones that incorporate transportation costs/time and accessibility into their models of how cities work.

Then, I want to establish a foundation for how infrastructure creates good mobility, and vice versa. What role do potholes have on travel times? How does street design affect accidents? Where do crashes tend to happen in cities? This literature will help connect the individual elements of my data to the broader concept of mobility friction.

Lastly, I want to shout out existing projects at this same intersection of data and mobility. NYC's Vision Zero work is an especially exciting example of how spatial data can be used to understand street safety. These projects provide both methodological inspiration and a comparison for what my project adds by thinking about mobility more broadly than crashes.

3. DATA

The project will combine several large and publicly available datasets from NYC Open Data: 311 service requests, motor vehicle collisions, and parking violations and more. This section will explain what the data is and who produces it (NYPD, DOT, and other city agencies), their timeframe (mostly 2020 to the present), and their scale (many millions of observations).

I will also explain the decisions I make to edit and limit the data. For example, the 311 dataset contains hundreds of complaint types, are outside the scope of the project. I am narrowing those down to complaints like street and sidewalk conditions, traffic signals, obstructions, and illegal parking, and will make similar decisions for the other datasets.

The data is, in itself, both accessible and clean. What makes my project interesting is not the difficulty of using the data, but the sources I am combining in an effort to think holistically about what it is like to travel a specific street. I will also discuss limitations, particularly that 311 measures what people report, rather than necessarily the true existence of a problem.

4. METHODOLOGY

The challenge of the project is turning individual observations into a meaningful story of a street. I plan to connect these observations from the different datasets spatially to streets and then summarize the types and frequency of friction occurring on each. Rather than treating every source of friction equally, I plan to divide them by mobility type: pedestrian, bicycle, car. A pedestrian is not concerned about the highway, whereas a car is not worried about the sidewalk. The product is an interactive dashboard where users can explore how mobility friction varies across both the city and different means of mobility.

This section will also explain how I produce summary statistics and make decisions about weighting. One challenge, for example, is distinguishing a street with lots of friction from a street that simply has lots of people using it. If the project develops far enough, I would also like to explore regressions or other statistical analyses of what characteristics predict mobility friction.

5. RESULTS

The results section will focus on the spatial trends that emerge from the data. Where has the most friction? How does friction vary by type? Do the streets that are difficult for drivers look different from those that are difficult for pedestrians or cyclists? How do those patterns vary across the city?

6. CONCLUSION

I will summarize the major spatial patterns, discuss what changes when mobility is considered separately by mobility type, and consider what combining these datasets adds to our understanding of NYC streets. Plus, a discussion of the limitations of the project and possible extensions because the resulting measures will reflect reporting, enforcement, and differences in how heavily streets are used, rather than perfectly capturing every traveler's experience.

7. SOURCES (so far)

D'Almeida, A. C. (Ed.). (2018). *Smarter New York City: How City Agencies Innovate*. Columbia University Press. http://www.jstor.org/stable/10.7312/dalm18374

Glaeser, E. (2011). Triumph of the city: How our greatest invention makes us richer, smarter, greener, healthier, and happier. Penguin Press.

Glaeser, E. L., & Gottlieb, J. D. (2009). The wealth of cities: Agglomeration economies and spatial equilibrium in the United States. Journal of Economic Literature, 47(4).

Shahum, L. (2017). Safe Streets: Insights on Vision Zero Policies from European Cities. German Marshall Fund of the United States. http://www.jstor.org/stable/resrep18873.